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MIL-I-25120C(USAF)
1 JUNE 1993
SUPERSEDING
MIL-I-25120B(USAF)
23 JULY 1974

MILITARY SPECIFICATION

INDICATOR, INSTRUMENT LANDING, AIRPORT RUNWAY

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electrical indicating instrument, designated Indicator Instrument Landing, Airport Runway, that is used to indicate the position of an aircraft relative to the end of a runway at an airport.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

L-P-513	Plastic-sheet, Laminated Thermosetting, Paper-Base, Phenolic-Resin

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610

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PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

Military

MIL-M-14	Molding Plastics And Molded Plastic Parts, Thermosetting
MIL-W-76	Wire And Cable, Hook-up, Electrical, Insulated
MIL-P-116	Preservation Methods Of
MIL-B-131	Barrier Materials, Watervaporproofed, Greaseproof, Flexible, Heat Sealable
MIL-T-152	Treatment, Moisture-And Fungus- Resistant Of Communications, Electronic And Associated Electrical Equipment
	Sheets, Glass Cloth, Silicone Resin
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use And Static Dehumidification
MIL-C-5015	Connectors, Electrical, "AN" Type Base, Phenolic-Resin
MIL-I-24768/1	Insulation, Plastics, Laminated, Thermosetting, Glass-cloth, Melamine Re Sin (GME)
MIL-I-24768/9	Insulation, Plastics, Laminated, Thermosetting Nylon-fabric-base, Pheno Lic-resin (NPG)
MIL-I-24768/17	Insulation, Plastics, Laminated, Thermosetting, Glass-cloth, Silicone-res In (GSG)
MIL-D-25177	

Federal

FED-STD-101	Test Procedures For Packaging Materials
FED-STD-595	Color

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-970	Standard & Specification, Order Of Preference For The Selection
MS3102	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, an Type
MIL-STD-810	Environment Test Methods and Engineering Guildelines
MIL-STD-2073/1B	DOD Materiel Procedures For Development & Application Of Packaging

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MS20003	Indicator, Humidity, Card, Three Spot, Impregnated Areas
MS25331	Light Assembly

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U. S. Air Force

45G40358, Rev. G. Indicator ID-48( )/ARN, Outline Dimensions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.2.1) in accordance with 4.4.

3.2 Material.

3.2.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.2.1.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they develop suitable properties and are replaceable by the An or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.2.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life. Materials contained within hermetically sealed enclosures are considered to be suitably protected from corrosion. Requirements specified for fungicidal and corrosion-protective treatment, anodizing of aluminum alloy parts, and cadmium plating of steel parts are not applicable for parts within hermetically sealed instruments.

3.2.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.2.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, they shall be treated with a fungicidal agent, as approved by the procuring activity.

3.2.5 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage; they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or with extremes of atmospheric conditions shall be avoided.

3.2.6 Plastic materials. Laminated thermosetting plastic materials shall be in accordance with the requirements of MIL-M-14, grade MTS-E-4, MIL-I-24768/17, L-P-513, type PBE or PBE-P, MIL-I-24768/1, or MIL-I-24768/9 or arc-resistant melamine types of laminated or molded materials.

3.2.7 Tropicalization. The indicator shall be tropicalized in accordance with MIL-T-152. All hookup wire shall comply with MIL-W-76. Moisture-and fungus-resistant plastic sleeving shall be used where sleeving is required. All waxes shall be eliminated where practicable; when their use is necessary, a high-grade microcrystalline wax containing a suitable fungicide, or a high-grade wax coated with moisture-and fungus-resistant material shall be used except where the component is hermetically sealed.

3.2.8 Reclaimed materials. This use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3 Design. The indicator shall be designed as a direct-current instrument which provides visual indication of aircraft position, by means of two crossed pointers. Two warning flag mechanisms and a marker beacon lamp will be provided.

3.3.1 Function. The vertical pointer shall indicate lateral deviation from the on-course signal of a selected runway localizer. The horizontal pointer shall indicate vertical deviation from an on-course glide slope. The marker beacon lamp shall provide visual indication of the marker beacon signals and the flag alarms shall provide nonoperational indication of the associated electronic equipment.

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3.4 Construction. The indicator shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service use.

3.5 Detail of components.

3.5.1 Indicator case. The indicator case shall conform to Drawing No. 45G40358, revision G.

3.5.2 Face design. The indicator face design shall be in accordance with Drawing No. 45G40358, revision G.

3.5.3 Cover glass. The cover glass shall be clear, flat, and free from flaws which interfere with normal reading of the indicator.

3.5.4 Vertical pointer. The vertical pointer shall be pivoted at the top, shall normally be vertical at center zero, and shall deflect right or left in agreement with the polarity of direct current received from a localizer receiver. The vertical pointer shall be positioned behind the horizontal pointer when facing the front of the indicator.

3.5.4.1 Deflection. The vertical pointer shall be designed to deflect to the right when terminals A and B of the receptacle specified herein are connected to a voltage which is polarized plus and minus, respectively.

3.5.4.2 Response. The vertical pointer response shall be linear in degrees with respect to current within 7.5 percent of the proportionate full-scale value. A deflection of five-eighths of an inch measured from the center of the dial to the pointer along the horizontal line of dots shall require 150 microamperes, plus or minus 5 percent, with the indicator in a normal upright position.

3.5.4.3 Resistance. The resistance across terminals A and B shall be 1000 ohms plus or minus 3 percent at 25°C (77°F).

3.5.4.4 Damping. The damping mechanism shall be over-damped (no over-swing).

3.5.4.5 Response time. Response time of the indicating pointers shall be between 0.80 and 1.15 seconds. Response time is defined as the time required for the pointer to reach 90 percent of its final indication, counted from the instant of application of the current. Final indication is defined as the point at which the indicating pointer has come to apparent rest when it has reached plus or minus 1 percent of the scale length of the actual rest point.

3.5.5 Vertical flag alarm. The vertical flag alarm shall be pivoted at the bottom, and shall be a suppressed zero class of mechanism, held against a stop by the suppression in a normal vertical position.

3.5.5.1 Deflection. The vertical flag alarm shall be designed to deflect

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clockwise when terminals E and F or the receptacle specified herein are connected to a voltage which is polarized plus and minus, respectively.

3.5.5.2 Response. The vertical flag alarm response shall be linear in degrees with respect to current. The alarm flag shall rest against its stop, with a current of 125 microamperes plus or minus 5 percent. The alarm flag shall be covered by the face mask, with a current of 250 microamperes plus or minus 5 percent.

3.5.5.3 Resistance. The resistance across terminals E and F shall be 1000 ohms plus or minus 3 percent at 25°C (77°F).

3.5.5.4 Damping and response time. The values of damping and response time are not critical but shall conform to good instrument practice.

3.5.6 Horizontal pointer. The horizontal pointer shall be pivoted at the left, shall normally be horizontal at center zero, and shall deflect up or down in agreement with the polarity of direct current received from a glide-slope receiver.

3.5.6.1 Deflection. The horizontal pointer shall be designed to deflect downward when terminals C and D of the receptacle specified herein are connected to a voltage which is polarized plus and minus respectively.

3.5.6.2 Response. The horizontal pointer response shall be linear in degrees with respect to current within 7.5 percent of the proportionate scale value. A deflection of five-eights of an inch measured from the center of the dial to the pointer along the vertical line of dots shall require 150 microamperes plus or minus 5 percent, with the indicator in a normal upright position.

3.5.6.3 Resistance. The resistance across terminals C and D shall be 1000 ohms plus or minus 3 percent.

3.5.6.4 Damping. The damping mechanism shall be over-damped (no over-swing).

3.5.6.5 Response time. Response time of the indicating pointers shall be between 0.80 and 1.15 seconds. Response time is defined as the time required for the point to reach 90 percent of its final indication, and time is counted from the instant of application of the current. Final indication is defined as the point at which the indicating pointer has come to apparent rest when it has reached plus or minus 1 percent of the scale length of the actual rest point.

3.5.7 Horizontal flag alarm. The horizontal flag alarm shall be pivoted at the right and shall be a suppressed zero class of mechanism, held against a stop by the suppression in a normal horizontal position.

3.5.7.1 Deflection. The horizontal flag alarm shall be designed to deflect clockwise when terminals G and H of the receptacle specified herein are connected to a voltage which is polarized plus and minus, respectively.

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3.5.7.2 Response. The horizontal flag alarm response shall be linear in degrees with respect to current. The alarm flag shall rest against its stop, with a current of 125 microamperes, plus or minus 5 percent. The alarm flag shall be covered by the face mask, with a current of 250 microamperes, plus or minus 5 percent.

3.5.7.3 Resistance. The resistance across terminals G and H shall be 1000 ohms plus or minus 3 percent at 25°C (77°F).

3.5.7.4 Damping and response time. The values of damping and response time are not critical but shall conform to good instrument practice.

3.5.8 Pointers. The visible part of the pointers shall not be less than 0.045 inch or more than 0.055 inch wide. The free ends of the pointers shall not be visible from directly in front of the indicator, regardless of the position of the pointers. The pivot ends of the pointers shall not project more than five-thirty-seconds inch from the edge of the dial mask measured parallel to the dial, and with the pointers in a center zero position.

3.5.9 Zero adjust. Each pointer mechanism shall be provided with an external zero adjustment.

3.5.10 Pointer stops. Pointer stops for both vertical and horizontal pointers shall be located beyond the dots located five-eighths of an inch from center so that it will be possible to measure full-scale deflection without the pointers being against the stops.

3.5.11 Connectors. The four meter movements shall be wired to a 10-terminal receptacle (MS3102-18-1P) mounted on the indicator. The lamp and switch assembly shall be wired to a 3-terminal receptacle (MS3102-14S-7P) mounted on the instrument. The receptacles shall be in accordance with MIL-C-5015.

3.5.12 Shielding. The indicator shall be magnetically and electrically shielded. When held in any position so that the nearest portion of the indicator is a distance of 8 inches from the pivot of the card of an aircraft compass in a horizontal magnetic field having a strength of 0.18 gauss, the indicator shall not cause a change in reading of the compass card of over 4 degrees.

3.5.13 Mounting. The indicator shall be designed for front mounting and each of the three mounting lugs of the indicator face shall withstand loads up to 150 pounds attached to screws in the mounting holes.

3.5.14 Parallax. The vertical pointer shall be placed next to the scale. The outside horizontal pointer shall not be farther than one-fourth inch from the scale. This distance shall be held to a practical minimum in order to reduce parallax. The distance between the front surface of the dial and the inner surface of the glass shall not exceed three-eighths inch.

3.5.15 Fixed reference marks. A projection of the vertical line of dots as

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shown on Drawing No. 45G40358 shall coincide with the dividing line between the yellow and blue sectors within 0.015 inch.

3.5.16 Interference. Rapidly energizing the marker lamp shall not product any noticeable effect on either of the two pointers.

3.5.17 Overload. The pointers shall not stick, tangle, or bend when either or both are subjected to steady or intermittent overloads up to four times full-scale current.

3.5.18 Operating position. The indicator shall operate correctly when mounted in any position.

3.5.19 Service conditions.

3.5.19.1 Vibration. Both pointers and flags shall be free from errors caused by vibration as experienced in aircraft use.

3.5.19.2 Barometric pressure. The indicator shall operate correctly over a range in barometric pressure from 3.5 to 30 inches of mercury; the pressure may remain approximately constant for long periods of time or may vary at a rate as high as 0.5 inch of mercury per second.

3.5.19.3 High temperature. The pointers shall not show evidence of sticking when operated at temperatures as high as plus 71°C.

3.5.19.4 Low temperature. The indicator shall provide the performance specified herein during and after subjection to temperatures as low as -55°C.

3.5.19.5 Relative humidity. The indicator shall be capable of withstanding, without damage, a relative humidity ranging up to 95 percent over long periods of time.

3.5.19.6 Nature of surroundings. The indicator shall be capable of operating under conditions of saltwater spray, hot engine gases, oil and water spray, or any other condition likely to occur when in use in aircraft.

3.5.20 Dial markings.

3.5.20.1 Scale. The scale of the indicator shall be dull black with luminescent markings as shown on Drawing No. 45G40358.

3.5.20.2 Localizer segment. The yellow fluorescent materials used on localized segments shall be in accordance with FED-STD-595, color No. 33907.

3.5.20.3 Flag alarm. The red fluorescent materials used on both pointer flag alarms shall be in accordance with FED-STD-595, Color No. 31905.

3.5.21 Dielectric strength. The indicator shall be capable of withstanding a voltage of 250 volts rms 60 cps ac between any moving coils and 500 volts rms 60

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cps ac between any moving coil and the case.

3.5.22 Marker beacon. Marker beacon lamp and switch assembly shall comply with MS25331.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The identification data applied to the indicator shall be as follows:

3.6.1 Marking. The manufacturer's code name, order number, and the indicator serial number shall be marked with contrasting black, or stamped on the face of the indicator as shown on Drawing No. 45G40358, so that only the pointers, alarm flags, color sectors, and fluorescent dial markings shall be visible in subdued light or in ultra-violet light.

3.7 Weight. The weight of the complete indicator shall not exceed 2.2 pounds.

3.8 Workmanship

3.8.1 General. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.8.2 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.8.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.8.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.8.5 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Acceptance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 in the specification.

4.4 First article inspection

4.4.1 First article test samples. When specified in the contract, preproduction test samples shall consist of 2 each Indicator ID-48( )/ARN and one complete set of indicator parts (unassembled) and shall be subjected to the preproduction tests specified herein. Samples shall be identified with the manufacturer's own part number and any additional information required by the authorizing letters.

4.4.2 Tests. The first article tests shall consist of the tests listed under acceptance tests and the following:

4.4.2.1 Electrical tests. The indicators shall be tested for suitability with respect to damping, resistance of indicators movements, and overload.

4.4.2.2 Environmental tests. The indicators shall be subjected to the following tests in accordance with MIL-STD-810C.

a. Temperature-altitude, Method 504-1, Procedure I, Equipment Category 5, included.
b. Fungus-resistance test Method 508.2.
c. Salt fog, Method 509.1.

4.5 Acceptance inspection. Acceptance tests shall consist of:

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a. Individual tests
b. Sampling plan and tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests:

4.5.1.1 Visual test. The indicator shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.5.1.2 Response tests. The response of the vertical pointer and of the horizontal pointer and the flag alarms shall be measured. After the 100 percent calibration has been established, the voltage source shall be removed and freedom of movement of the pointers observed to assure that there is no sticking of the pointers to the meter movement stops.

4.5.1.3 Dielectric strength. The indicator shall be subjected to a voltage of 250 volts rms, 60 cps ac between the moving coils, and 500 volts rms, 60 cps ac between either moving coil and the case.

4.5.2 Sampling plan

4.5.2.1 Sampling tests. Two percent of the indicators, but not less than two indicators, shall be selected at random from the total number of indicators on order and subjected to the sampling tests.

4.5.2.2 Cover glass. The cover glass shall be subjected to the following test: A piece of tool steel 0.640 of an inch long, 0.262 of an inch in diameter, and rounded at the ends with a radius of approximately 0.160 of an inch shall be dropped endwise from a distance of 18 inches to the center of the cover glass. There shall be no evidence of cracking or shattering.

4.5.2.3 Vibration and temperature cycle. The equipment shall be vibrated with the frequency varying between 10 and 30 cycles per second at a total excursion of 1/32 of an inch, the frequency varying uniformly from 10 to 30 cycles per second and returning to 10 cycles per second in approximately 1/2 minute. Immediately following vibration in each direction, the indicator shall be subjected to a temperature cycle consisting of exposure to minus 55°C for 1 hour and to plus 71°C for 1 hour and return to room temperature. There shall be no evidence of sticking of either pointer or flags during or after this temperature cycle. Application of the vibration in the above test shall be in each of the following directions:

a. Horizontally, parallel to the major horizontal axis, for a period of 15 minutes.
b. Horizontally, at right angles to the major horizontal axis, for a period of 15 minutes.
c. Vertically, for a period of 15 minutes.

4.5.2.4 Humidity test. The indicator shall be placed in a humidity chamber and maintained at 50°C and 95 percent relative humidity for 96 hours. The indicator

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shall be maintained at two-thirds full scale deflection during this period. The indicator shall operate satisfactorily upon return to room temperature and shall show no evidence of corrosion.

4.6 Acceptance criteria. When one or more items fail to meet the specification, acceptance of all items will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.7 Packaging inspection

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection and rough handling tests, as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (see 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in quantity unit pack (QUP) of one each. Each item will be provided a preservation method IIc in accordance with (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073/1B to provide a method IIc.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073/1B.

5.2 Packing. Packing shall be level "A", "B", "C", as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-

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B-131, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material between the item and unit container to protect the unit container from punctures. Intermediate container requirements shall conform to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Intermediate containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant, unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, SPI, and contracting activity.

5.2.3 Level B. Same as Level "A" unless otherwise specified by SPI or the contracting activity.

5.2.4 Level C. Item will be packed in containers in such a manner as to afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073/1B.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contractor, SPI and MIL-STD-129.

5.4 Inspection and Test. Test of methods of preservation shall be accomplished IAW section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Indicator ID-48( )/ARN covered by this specification is a crossed-pointer type instrument providing indication form the localizer receiving equipment, from the glidescope receiving equipment, or from a combination of the two. Indication is also provided for the marker beacon receiver. Flag alarms provide a warning of nonoperation of receiving equipment or ground transmitting equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
c. When first article is required (see 3.1).
d. Level of preservation and packing (see 5.1 and 5.2).

6.2.1 First article. It is expected that the contract of purchase order will specify that 2 each Indication ID-48( )/ARN and one complete set of indicator parts (unassembled) will be required for preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of this specification. These samples should be considered an additional quantity since it may not be economical to repair and deliver these indicators as production items. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Interchangeability. It is the intent of this specification to provide an indicator that is equal to and interchangeable with existing indicators procured by previous editions of this document. Refer to Technical Order 12R5-2ARN-144 for detailed parts list.

6.4 Subject term (key word) listing.

Glideslope
Localizer
Marker beacon

6.5 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this standard is as follows: NONE

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing activity:
AIR FORCE -71

Project number:
6610-F356

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-25120C	930601

3. DOCUMENT TITLE

INDICATOR, INSTRUMENT LANDING, AIRPORT RUNWAY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
HARLENA EDWARDS		336-5960
c. ESS (Include Zip Code) OC-ALC/TICLA TINKER AFB, OK 73145-3037	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289 2340	

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